

Turning casual rides into annual memberships

Bike-share behavior analysis using BigQuery SQL and Tableau

5.87M

cleaned rides

12 months of trip data

The business question is conversion—not just usage

How do annual members and casual riders use Cyclistic bikes differently?

Decision to support

Identify practical marketing actions that can encourage casual riders to become annual members.

Analytical lens

Compare ride volume, duration, weekday, month, hour, bike type, and start-station behavior.

The goal: translate behavior patterns into testable conversion tactics.

Twelve monthly files became one analysis-ready table

01

Collect

12 monthly CSV
files
Aug 2025–Jul 2026

02

Validate

Schema, row
counts,
missing values,
duplicates

03

Clean

Remove duplicate
IDs and
invalid ride
durations

04

Analyze

SQL aggregations
and
Tableau dashboard

Tools

Google BigQuery / SQL • Tableau Public • GitHub

Data quality checks removed invalid records before analysis

6,037,933

initial rows



5,871,668

final cleaned rides

35

duplicate ride IDs identified and removed

< 1 min

rides excluded as too short

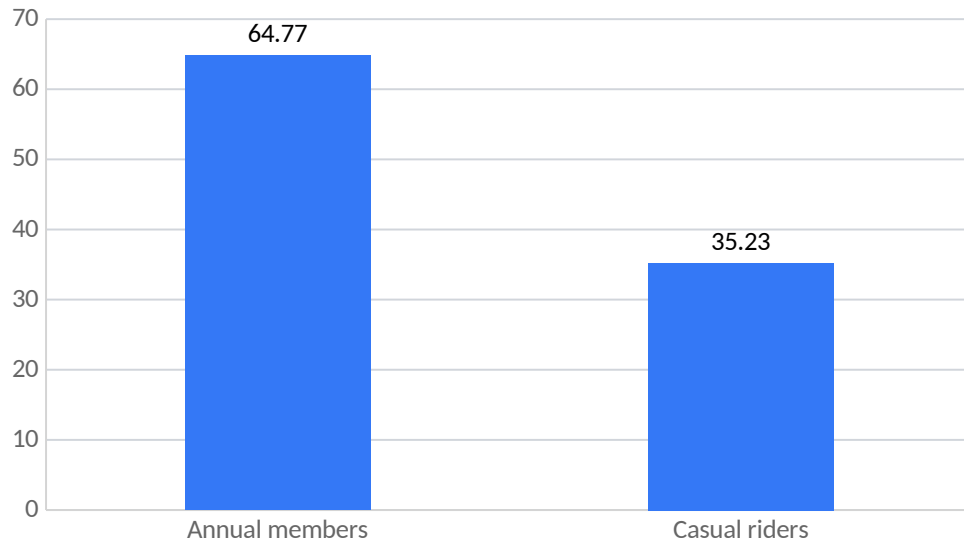
≥ 24 h

rides excluded as implausibly long

Result: a validated dataset suitable for reliable behavioral comparison.

Members drive more rides; casual riders stay longer

Share of all rides



Members account for 64.77% of rides.

Average ride duration



Casual trips are 54% longer on average.

The two groups use Cyclistic for different jobs

ANNUAL MEMBERS

Routine transportation

- Stronger weekday activity
- Clear commuting-hour peaks
- Shorter, more frequent rides

CASUAL RIDERS

Leisure and recreation

- Stronger weekend activity
- More rides later in the day
- Longer average trips

Implication: conversion messages should match the moment and purpose of the casual ride.

Five conversion actions follow directly from the evidence

1	Seasonal membership campaign	Promote before and during spring–summer demand; test a trial or limited-time discount.
2	Weekend and leisure targeting	Reach casual riders when they are most active and connect leisure use with membership convenience.
3	Show the financial value	Compare repeated single-ride or day-pass spending with the annual membership price.
4	Activate high-casual stations	Use QR codes, local ads, and membership offers at stations favored by casual riders.
5	Test and measure conversion	A/B test messages and offers; track conversion, CTR, cost per conversion, and repeat rides.

Make membership relevant at the right moment

Casual riders already show strong engagement—especially on weekends, later hours, and longer trips.

Recommended next step

Run a targeted pilot campaign at high-casual stations during warm-season weekends, then measure membership conversion against a control group.

Data limitation

Trip data does not include customer demographics, pricing history, or individual purchase behavior.

Portfolio links

[GitHub repository](#) • [Interactive Tableau dashboard linked in the repository](#)